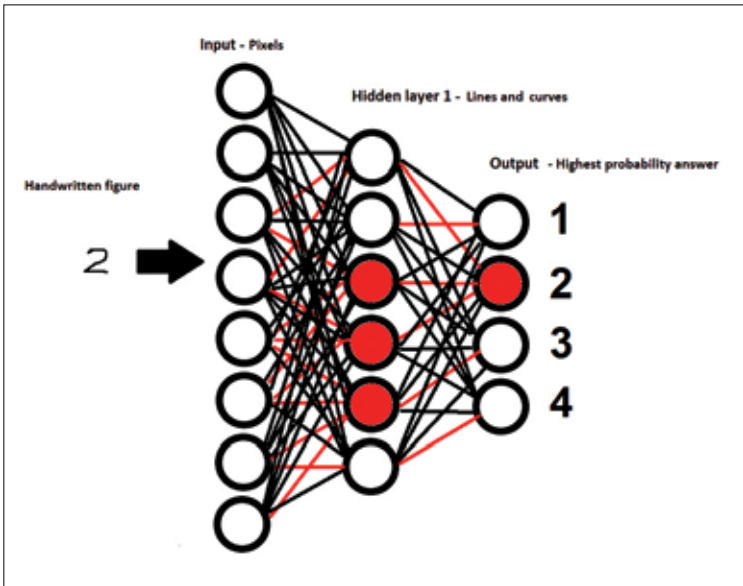




Deep neural networks comprise of multiple hidden layers.

of hidden layers, typically more than three, 'deep' within the surface of the original input and output layers.

Deep learning is especially useful in complex tasks like understanding the value of a handwritten digit or alphabet in various writing styles. With a population of over 7.7 billion, the world has countless different ways of writing a digit or alphabet. Gauging the value of a digit consistently through various different styles requires deep learning since the invisible neural layers allow enhanced interconnectedness. This, in turn, allows the comparisons and predictions to be more accurate than simple input and output method.

Multiple neural layers in deep learning allow a computer to provide a method of the neural network which can build a rough hierarchy or gradation scale of the various different features that make up, say a handwritten numerical digit. An example to show how multiple hidden layers work in analysing a handwritten digit is - the first layer relates the exact pixels in the image of the handwritten digit, the next layer combines these pixels to make lines, the next takes it further by assessing and forming shapes that correspond to the shape of the digit, next layer combines these shapes